Shuhao Zhang

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Education

ShanghaiTech University 2020.09 – 2024.07
Computer Science and Technology B.S. School of Information Science and Technology Shanghai

ShanghaiTech University 2024.09 – Present
Electronic Information (CS Track) Master School of Information Science and Technology Shanghai

Publications

S. Zhang, J. Dong, H. Wang, C. Jiang, and Q. Li. *When Seconds Count: Designing Real-Time VR Interventions for Stress Inoculation Training in Novice Physicians*. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 25 pages. https://doi.org/10.1145/3772318.3790470.

Y. Shao, Y. Xu, Y. Jin, **S. Zhang**, W. Gu, and Q. Li. *DesignBridge: Bridging Designer Expertise and User Preferences through AI-Enhanced Co-Design for Fashion*. In 31st International Conference on Intelligent User Interfaces (IUI '26), March 23 – 26, 2026, Paphos, Cyprus. ACM, New York, NY, USA, 27 pages. https://doi.org/10.1145/3742413.3789081.

X. Dou, J. Dong, **S. Zhang**, Q. Zhu, and Q. Li. *Danmaku Avatar: Enabling Asynchronous Co-viewing Experiences in Virtual Reality via Danmaku*. In the ACM CHI conference on Human Factors in Computing Systems (CHI) 2025 LBW Track.

H. Jiang, S. Shi, **S. Zhang**, J. Zheng, and Q. Li. *SLInterpreter: An Exploratory and Iterative Human-AI Collaborative System for GNN-based Synthetic Lethal Prediction*. in IEEE Transactions on Visualization and Computer Graphics, vol. 31, no. 1, pp. 919-929, Jan. 2025, doi: 10.1109/TVCG.2024.3456325 (Proc. IEEE VIS 2024).

Y. Xu, **S. Zhang**, Y. Fan, S. Shi, Z. Peng, and Q. Li. *ReviseMate: Exploring Contextual Support for Digesting Academic Paper Reviews*, in Proc. ACM Hum.-Comput. Interact., Vol. 9, No. 7, Article 321. Publication date: November 2025. https://doi.org/10.1145/3757502

Internship Experience

ViSeer LAB, ShanghaiTech University - Research Intern (Advisor: Prof. Quan Li) 2023.04 – Present

Since joining ViSeer Lab, I have actively participated in academic research and top-tier conference submissions, continuously improving my system development and academic writing skills while building close collaborations with my team and advisor. After earning my bachelor's degree, I continued my master's studies in the lab, focusing on the interaction design and development of VR/MR medical training systems. During this period, I led the development of a VR surgical simulation system based on Unreal Engine and Unity. Currently, I am developing a high-fidelity MR first-aid training project with haptic feedback, utilizing VIVE Focus Vision hardware and a custom-modified CPR manikin. These experiences have not only solidified my hardware-software co-development skills but also comprehensively enhanced my project management and engineering practice capabilities.

Epitome Inc. - Software Development Intern, Technical Department 2024.12 – 2025.12

During my one-year internship, I was primarily responsible for project development based on Unity and UE5 engines, deeply involved in writing core business logic and implementing interactive systems. I utilized C# to develop low-level logic such as sequence triggers and object interactions, and participated in system optimization regarding VR templates and hand-tracking features. Throughout the internship, I strictly adhered to company regulations, quickly familiarized myself with business processes, and successfully applied my knowledge in HCI and visualization to practical workflows. I demonstrated strong communication skills, teamwork, and solid professional fundamentals.

Research Experience

Paper Submission *When Seconds Count: Designing Real-Time VR Interventions for Stress Inoculation Training in Novice Physicians* 2025.06 – 2025.09

Co-First Author Accepted by ACM CHI 2026 (Top-tier Conference in HCI)

To address the decline in decision-making capabilities of novice physicians due to acute cognitive overload in surgical emergencies, I was deeply involved in the entire R&D process of a novel VR Stress Inoculation Training (VR-SIT) system. In the early stages, I led a formative study with 12 participants to identify novice physicians' core needs for immediate assistance, extracting three key intervention strategies: self-regulation assistance, procedural guidance, and emotional/sensory support. Subsequently, I designed and implemented the VR-SIT system integrated with a Just-In-Time Adaptive Intervention framework, which dynamically customizes support based on learners' cognitive and emotional states. The effectiveness of the strategies was validated through a user study involving 26 participants. As co-first author, I was responsible for advancing the overall process, independently drafted the main content of the paper, and led the observational study and subsequent user experiments.

Paper Submission *DesignBridge: Bridging Designer Expertise and User Preferences through AI-Enhanced Co-Design for Fashion* 2024.12 – 2025.04

Third Author Accepted by ACM IUI 2026

To address long-standing challenges in fashion design co-creation—such as low engagement, inefficient preference collection, and difficulties in feedback integration—I participated in developing a multi-platform AI-enhanced interactive system named DesignBridge. Based on insights from a formative study, the system constructs a three-stage collaborative workflow (initial design framing, preference expression collection, and preference integration design) to effectively bridge designers' expertise with real user preferences. I was primarily responsible for building the overall client-side architecture and implementing frontend functionalities, developing interactive tools that support intuitive preference expression to ensure efficient data collection. Additionally, I was deeply involved in the system evaluation phase, helping organize user experiments and validate the system's significant advantages in enhancing preference collection and assisting AI in professional design integration.

Paper Submission *SLInterpreter: An Exploratory and Iterative Human-AI Collaborative System for GNN-based Synthetic Lethal Prediction* 2024.01 – 2024.04

Third Author Accepted by IEEE VIS 2024 (Top-tier Conference in Visualization)

To meet domain experts' needs for exploring the predictive mechanisms of synthetic lethal relationships, I participated in proposing a two-stage human-AI collaboration framework. The first stage supports experts in optimizing data selection based on knowledge graphs to derive insights; the second stage ensures that continuously generated AI explanation paths closely follow realistic biological principles. I independently designed and built the human-AI collaborative visual analytics system, and was responsible for the training and parameter tuning of the Graph Neural Network (GNN) based prediction model. I played a major role in backend development and paper drafting, assisted the first author in writing the observational study and system implementation sections, and was deeply involved in method evaluation experiments, case studies, and expert user interviews.

Paper Submission *ReviseMate: Exploring Contextual Support for Digesting Academic Paper Reviews* 2023.07 – 2023.09

Second Author Accepted by ACM CSCW 2025 (Top-tier Conference in HCI)

To tackle challenges in digesting academic peer review feedback, such as high time consumption and reading fatigue, I participated in developing an HCI system named ”ReviseMate”. In the initial phase, I formulated research questions through user interviews and storyboarding, and subsequently undertook the core design and the majority of the code implementation. The system was ultimately proven superior to traditional baseline methods through a controlled user study with 31 participants and a real-world deployment with 6 participants. I assisted the first author in completing the controlled user study, field deployment, and drafting relevant sections of the paper.

Competition *ChinaVis 2023 Data Challenge (***Third Prize, Top 25%***)* 2023.05 – 2023.06

This project aimed to extract four main dimensions reflecting driving styles from raw traffic data and construct big data tags for profiling civilian drivers. I was primarily responsible for cleaning, processing, and modeling 500MB of plain text backend data, and participated in establishing the driver style evaluation criteria. Meanwhile, I applied the t-SNE method for dimensionality reduction and clustering, and utilized visual analytics methods to interpret, analyze, and evaluate the clustering results.

Skills & Summary

- **Tech Stack**: Proficient in Python, C, C++, C#, JavaScript; Familiar with Unity and Unreal Engine (UE5) development; Proficient in frontend visualization technologies such as D3.js.
- **Expertise**: Human-Computer Interaction (HCI), Data Visualization, Virtual Reality (VR) and Mixed Reality (MR) System Development.
- **Languages**: Proficient in English (CET-4: 629, CET-6: 530), with excellent capabilities in reading academic literature and writing high-level academic papers.
- **Self-Summary**: Solid foundation in computer science with rich engineering and research practice experience. Familiar with the complete project lifecycle, from user requirement research, system design, and code implementation to controlled user experiments. Strong self-motivation and fast learning ability, capable of effectively translating cutting-edge academic theories into engineering solutions for practical business problems. I look forward to continuously delivering value in industrial roles and further enhancing my system-level development and teamwork skills.